

Online Learning Platform Project

1. Project Overview

Students will build a mini online learning platform where users can browse courses, enroll, track progress, and simulate completing lessons. This project introduces interactivity, data handling, and user experience design

2. Core Features

Beginner Scope

- Homepage with featured courses
- Course listing page with title, description, and duration
- Individual course detail page
- Enroll button to add courses to "My Courses"

Intermediate Scope

- "My Courses" dashboard showing enrolled courses
- Progress tracking with percentage completion
- Lesson completion simulation (checkboxes or progress bar)
- Responsive design for mobile and desktop
- LocalStorage to persist enrolled courses and progress

3. Workflow & Milestones

- **Phase 1:** Setup folder structure and mock course data
- **Phase 2:** Build course listing and detail pages
- **Phase 3:** Implement enrollment system and "My Courses" dashboard
- **Phase 4:** Add progress tracking and lesson completion simulation
- **Phase 5:** Enhance with search/filter, responsive design, and persistent storage

4. Suggested File Structure

Organize files into folders for CSS, JS, images, and separate HTML pages for index, courses, course detail, and dashboard. You can use the Spring mvc to integrate all of these.

5. Learning Outcomes

- Handling Backend for dynamic content
- DOM manipulation and event handling
- LocalStorage for persistence

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- Responsive design principles
- Modular file organization (SPRING FRAMEWORK)

6. Stretch Goals (Optional)

- User login/signup (mock authentication)
- Certificate generation after course completion
- Admin panel to add/edit courses
- Integration with a video player for lessons

Tip for Students: Start small with static pages and basic enrollment, then gradually add progress tracking and persistence. This way, you'll see how a simple idea grows into a functional platform.